

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Weekly Progress Report — Week 1

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1 Introduction and Motivation

Fabric manufacturers will have to test each roll of fabric for surface quality like holes, stains, broken threads and irregular weaving. This inspection is done in a traditional way, manually, which can be time-consuming, tedious, labor intensive and susceptible to human error, especially when operated at high speeds.

The objective of this project is to have a computer-based system that can determine whether a defect exists on a fabric image and automatically analyzing the defect. Describe and, if available, pinpoint where it is.

For this course project we intentionally exclude deep learning and neural networks. Rather, the system will be developed completely based on classical technique. Digital Image Processing (DIP) techniques like filtering, texture and brightness, contrast and hue-saturation were used in the study. Analysis, thresholding and morphological operations. This approach maintains the system's explainability, and clearly illustrates the concepts of DIP

2 Problem Statement and Research Question

The system is provided with a grayscale image of a woven fabric patch.

- Identify if there is a fault in the fabric.
- Recognize and locate the problem area in the image at the pixel level.
- Give an indication of the certainty of the detection result.
- Use only classical DIP techniques to do the full process.

The project also includes the comparison of two approaches to selecting texture-analysis parameters. In the first approach, a set of predetermined Parameters for all fabric types, second automatically adjusts. the parameters in line with the particular properties of every textile.

Central Research Question: How does automatically adapting to the current situation affect the efficiency of the extraction process? The texture analysis parameters of different types of fabric are improved by substituting. detection and localization in comparison to a fixed set of parameters?

3 Objectives for the Semester

The overall goals of the project are:

- Gain solid background in classical image-processing. Techniques that are applicable to texture-based defect detection.
- Design and implement a complete classical DIP pipeline for Defects detection and localization.
- Compare and contrast fixed and adaptive parameter methods.
- Analyze the last system with basic quantitative methods performance measures.

4 Project and Dataset Setup

4.1 Repository Creation

A shared GitHub repository was set up, according to folder structure given for the course. The repository all four of the modules are included: reports, results, and code are all separate. Team members were added

as collaborators. It will be used for weekly Submissions and mid-semester and end-of-semester project deliverables.

4.2 Dataset Acquisition

We have been given **AITEX Fabric Image Database**, a publicly available dataset specifically developed for fabric-defect detection research. The dataset provides both normal and defective fabric images, along with pixel-level reference masks for the defective samples.

The dataset contains **245 images representing 7 different fabric structures**. Of these, **140 images are defect-free**, with 20 normal images available for each fabric type. The remaining **105 images contain fabric defects**. Each image has a resolution of **4096 × 256 pixels**, providing a large spatial area for texture analysis and defect localization.

5 Initial Literature Review

5.1 Texture Analysis

Texture analysis describes local visual patterns such as smoothness, roughness, regularity, and repetition. Since woven fabrics have characteristic repetitive thread patterns, defects such as holes, stains, and broken threads create local disturbances in the texture. Therefore, texture analysis provides a suitable basis for fabric-defect detection.

5.2 Local Binary Patterns (LBP)

The first texture descriptor studied was **Local Binary Patterns (LBP)**. LBP compares the intensity of neighboring pixels with the center pixel and represents these comparisons as a binary pattern. The distribution of these patterns can then be used to describe the texture of an image region. LBP was selected as the starting technique because it provides a simple and effective way to characterize local texture and serves as a foundation for studying the other texture descriptors planned for the project.

6 Work Completed This Week

The following tasks were completed during Week 1:

- Finalized the project topic, problem statement, and central research question.
- Created and configured the shared GitHub repository.
- Identified, downloaded, and extracted the AITEX fabric image dataset.
- Studied the fundamentals of texture analysis and its application to fabric-defect detection.
- Studied the Local Binary Patterns (LBP) texture descriptor.

Implementation Status:

No code has been written yet. Week 1 focused on project setup, dataset acquisition, literature review, and building the theoretical foundation for implementation.

7 Plan for Next Week

- Study the other two texture-analysis techniques.
- Look at the structure of the AITEX dataset, categories of fabrics and Describe in detail the different types of defects.
- Get details of the images and their reference/ground truth.